

# Visual Perception & Cognition

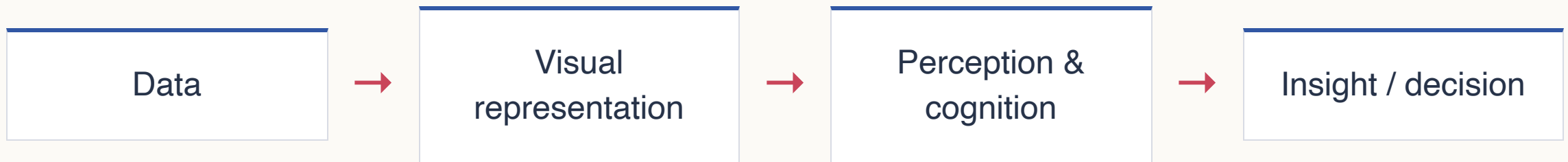
Designing for the mind behind the eyes

**Xianbin Gu · Lecture 2**



You can see everything—but what did you actually notice?

# Visualization works through a human “processor”



A design succeeds only when people can **notice**, **organize**, and **interpret** what matters.

# Today: from seeing to designing

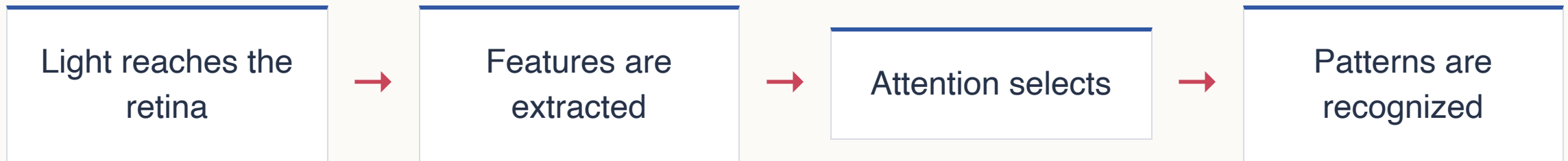
- |    |                                    |                                                     |
|----|------------------------------------|-----------------------------------------------------|
| 01 | <b>Seeing is active</b>            | Attention selects; knowledge guides interpretation. |
| 02 | <b>The mind organizes</b>          | Gestalt cues create groups, paths, and objects.     |
| 03 | <b>Principles become decisions</b> | Notice, group, follow, and interpret.               |
| 04 | <b>Bridge to encoding</b>          | Preview marks and channels.                         |

Part 01

# Seeing is active

The visual system filters and organizes before conscious reasoning begins.

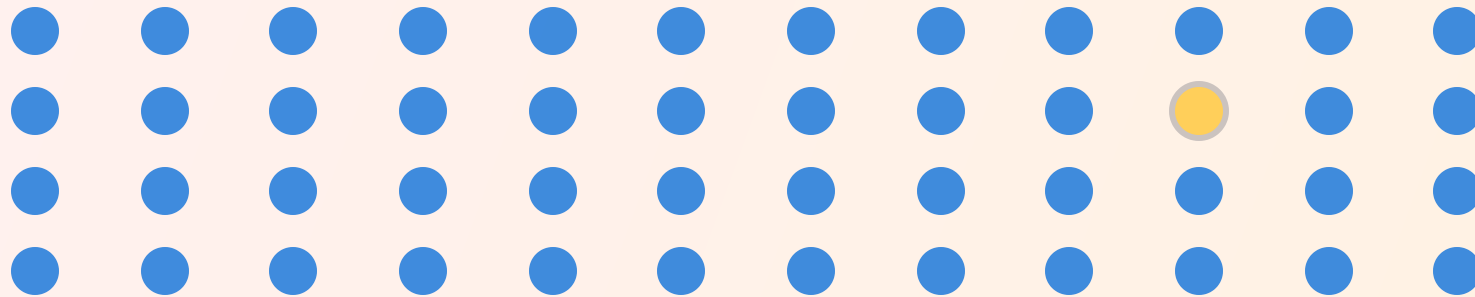
# Your eyes are not a camera



Edges, contrast, color, orientation, and motion are processed—not copied as an untouched picture.

QUICK SCAN · 5 SECONDS

# Find the one item that is different



# Some visual differences guide attention automatically

## Bottom-up

Driven by the stimulus

- contrast
- color difference
- orientation
- motion

## Top-down

Driven by the viewer

- goal
- expectation
- prior knowledge
- current task



# The same scene supports different searches



Your task changes what you see

Search for the person. Now, search for the equations.

The stimulus is unchanged; the goal redirects attention.

# What animals can you find?



Individual scan · 30 seconds · Then compare with a partner

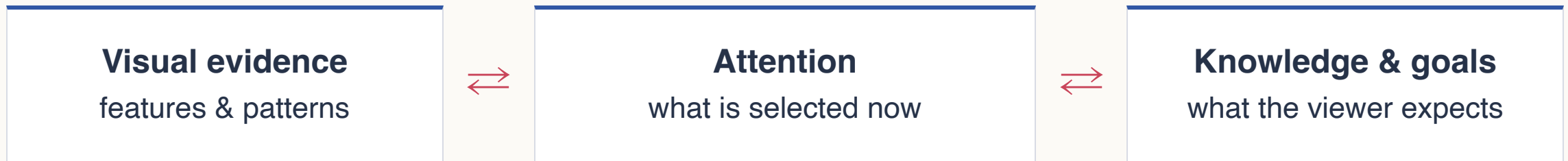
# Prior knowledge can dominate interpretation



## What do you see first?

Once one interpretation is suggested, it becomes difficult to “unsee.” Perception and cognition continually constrain one another.

# Perception and cognition form a loop



Good visualization supports the loop; misleading visualization exploits it.

# Perception and cognition form a loop



**Good visualization supports the loop**

The shape of the steering wheel reminds you the brand of the car.



# Perception and cognition form a loop



**Misleading visualization  
exploits it**

William Ely Hill, My wife and my mother-in-law, 1915

## WHY VISUALIZATION HELPS

Visualization is about external cognition: using resources outside the mind to increase what the mind can do.

Paraphrased from Stuart K. Card

- External resources become useful evidence only when the visual system organizes separate marks into meaningful groups, paths, and figures. Gestalt principles explain how that organization happens.

Part 02

# The mind organizes

Gestalt principles explain why separate marks become groups, paths, and objects.



# We perceive organized wholes—not isolated marks

## Without structure

Every item competes as an independent object.

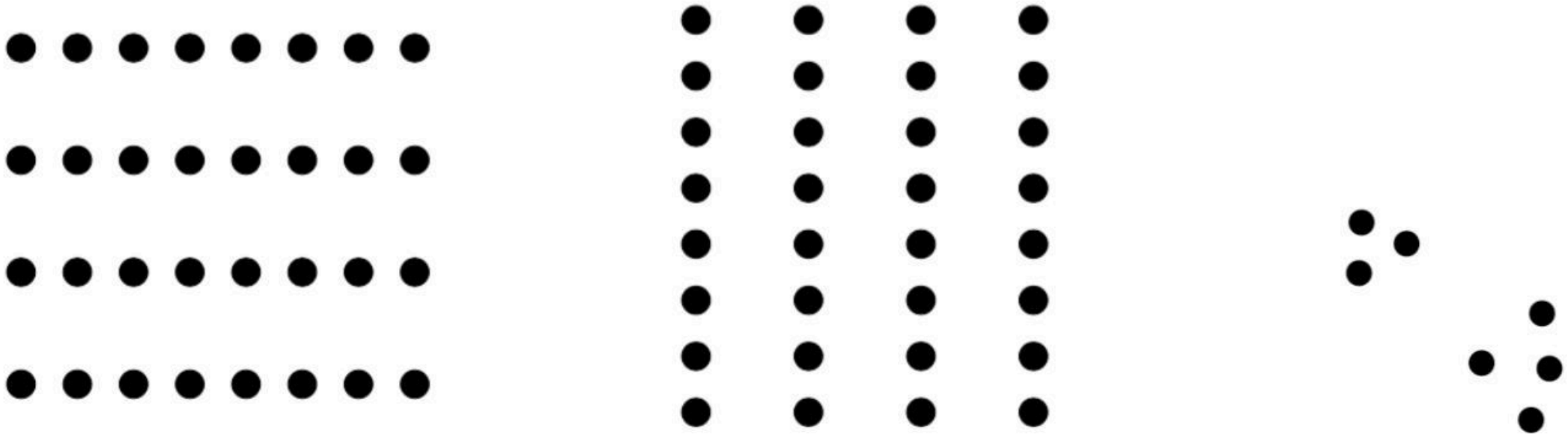


## With structure

Relations let the mind compress many items into a few groups.

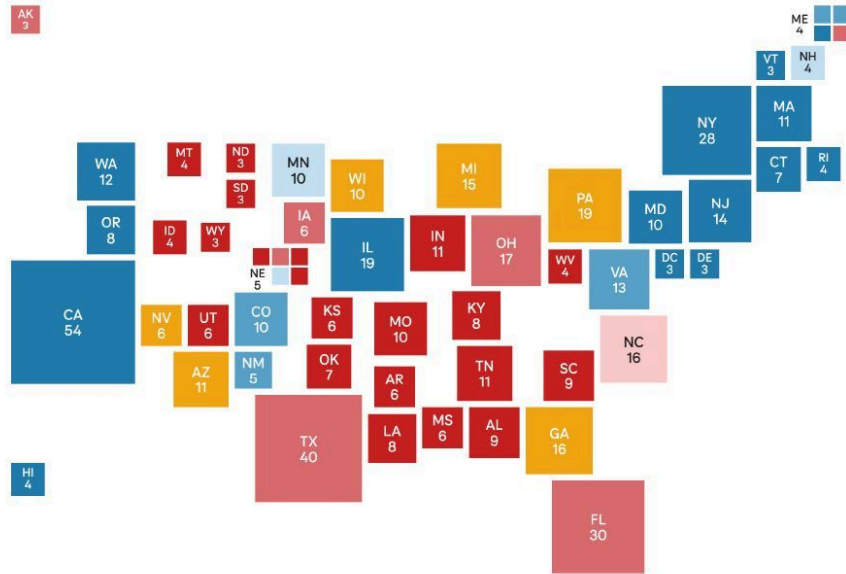


# Proximity makes groups before labels do



Use whitespace deliberately: distance is not empty—it encodes relationship.

# Using proximity in design



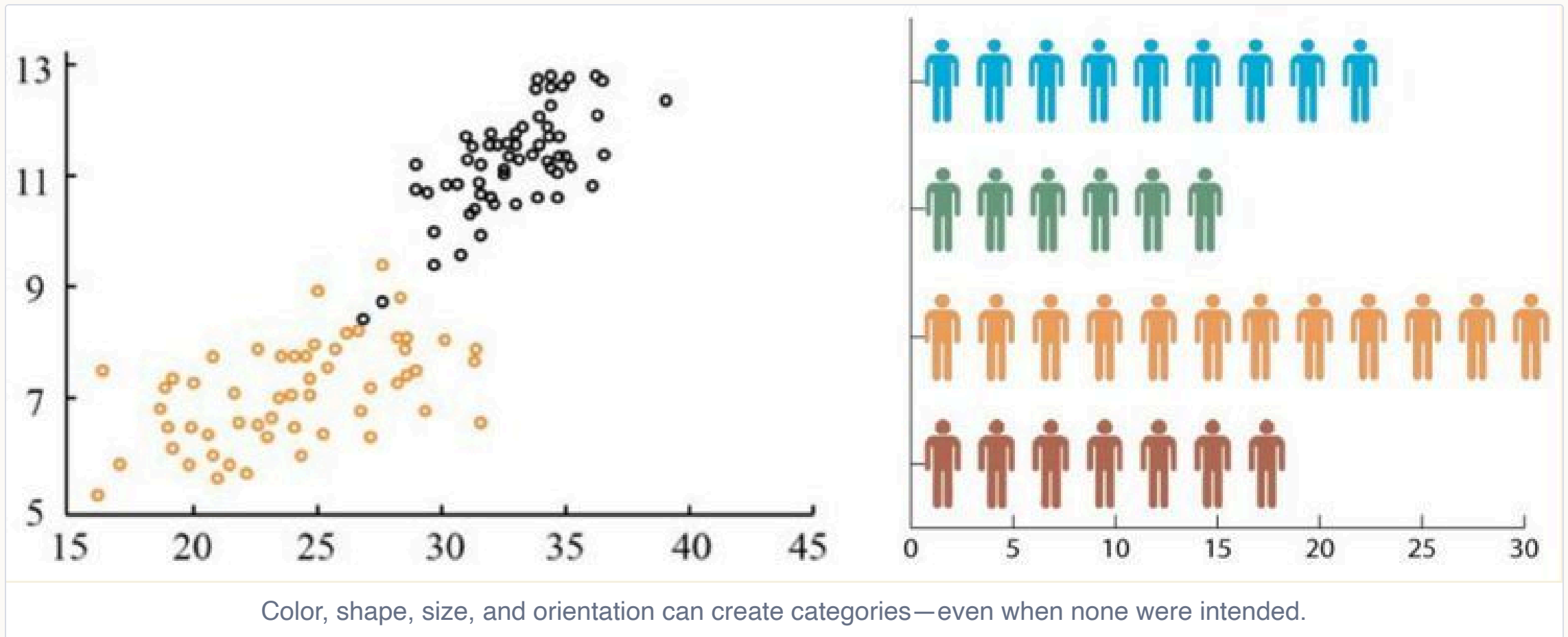
Nearby tiles form a recognizable map



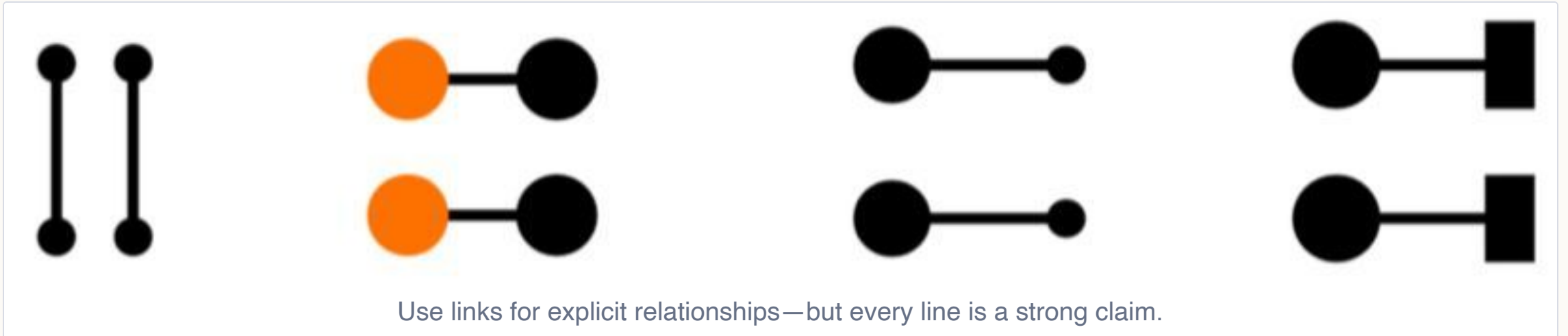
Nearby symbols form a unified logo

**Proximity creates structure.** Spacing helps viewers perceive separate elements as meaningful groups and larger forms.

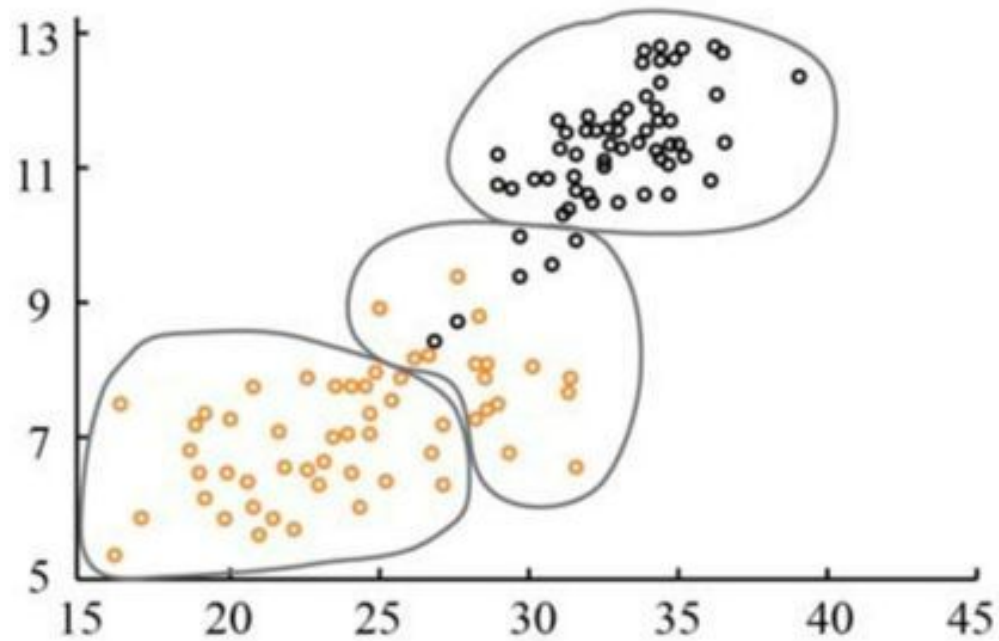
# Similarity says “these belong together”



# Connectedness can override proximity and similarity

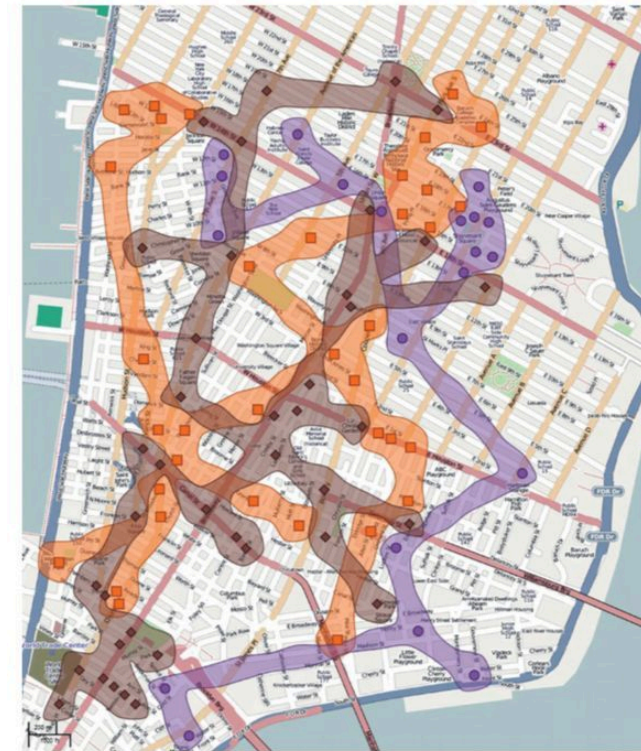


# Enclosure turns boundaries into groups



## Enclose related elements

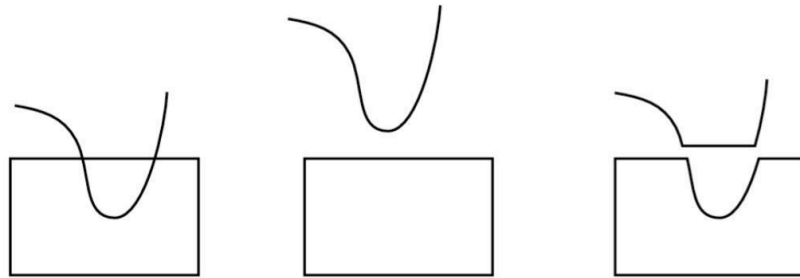
Elements within the same boundary are perceived as a group.



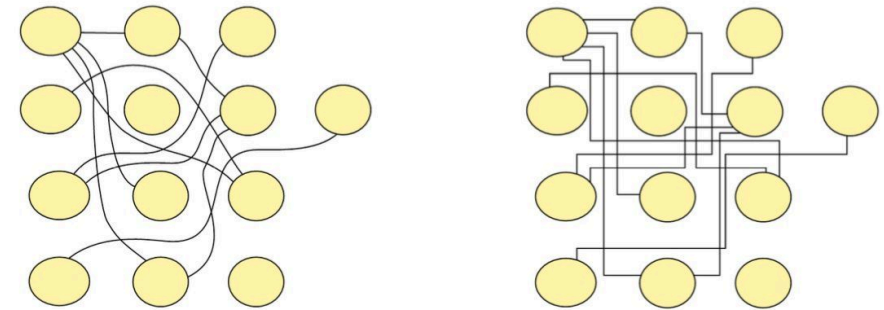
## Use regions to reveal groups

Contours can make spatial distributions readable.

# Continuity helps the eye follow a path

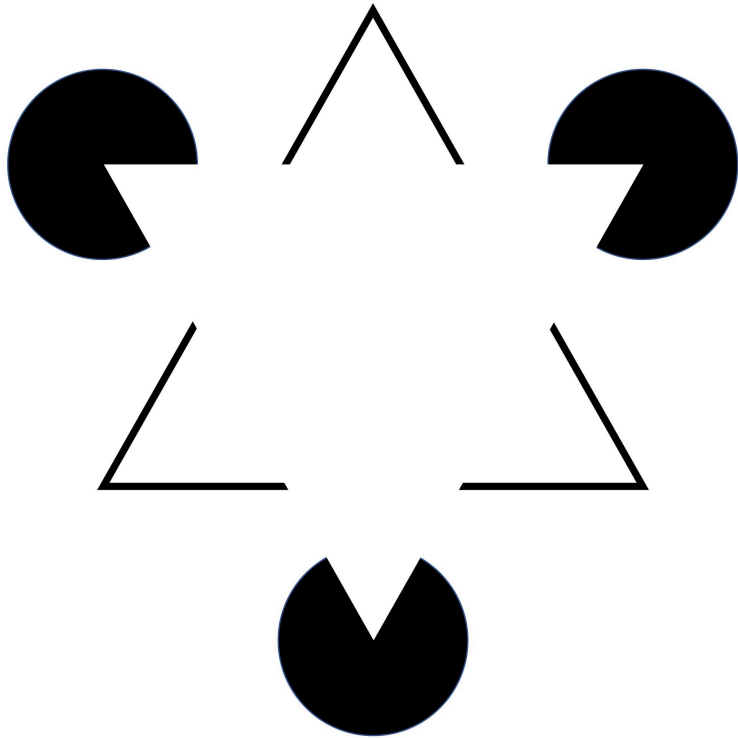


**Continuous boundaries are perceived as a whole.**

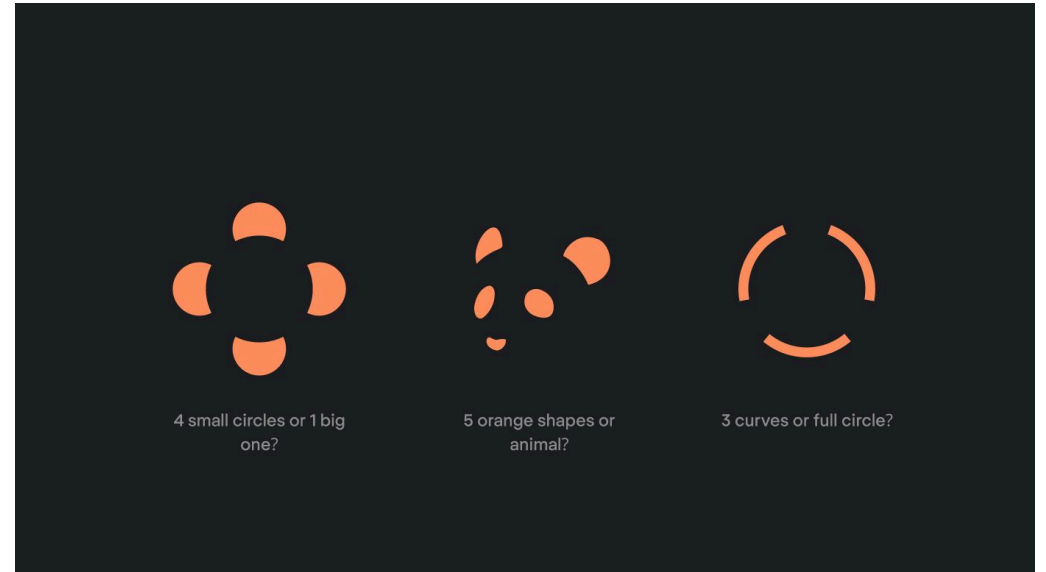


**Smooth paths are easier to follow. Abrupt turns make paths harder to trace.**

# Closure completes incomplete objects



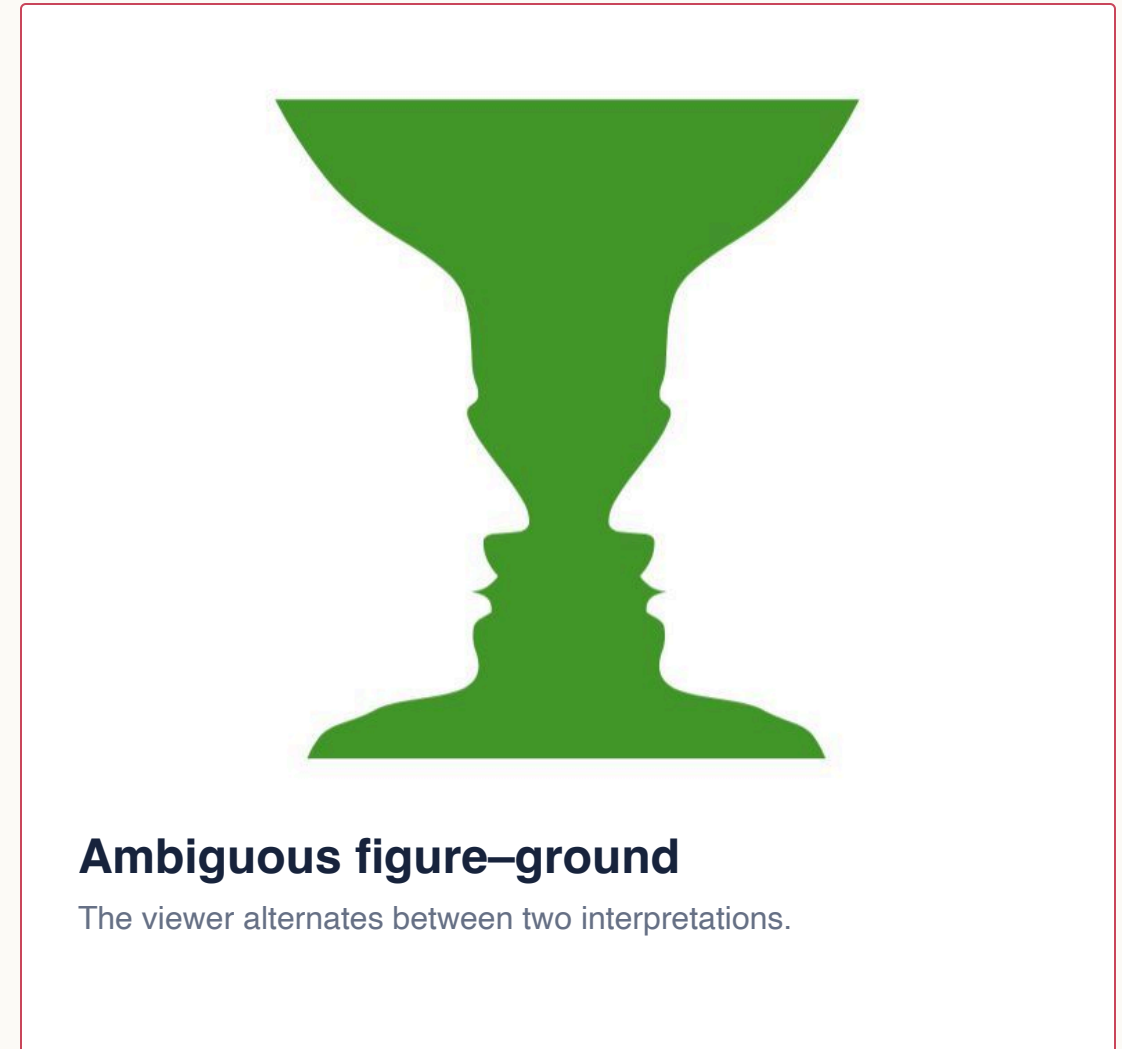
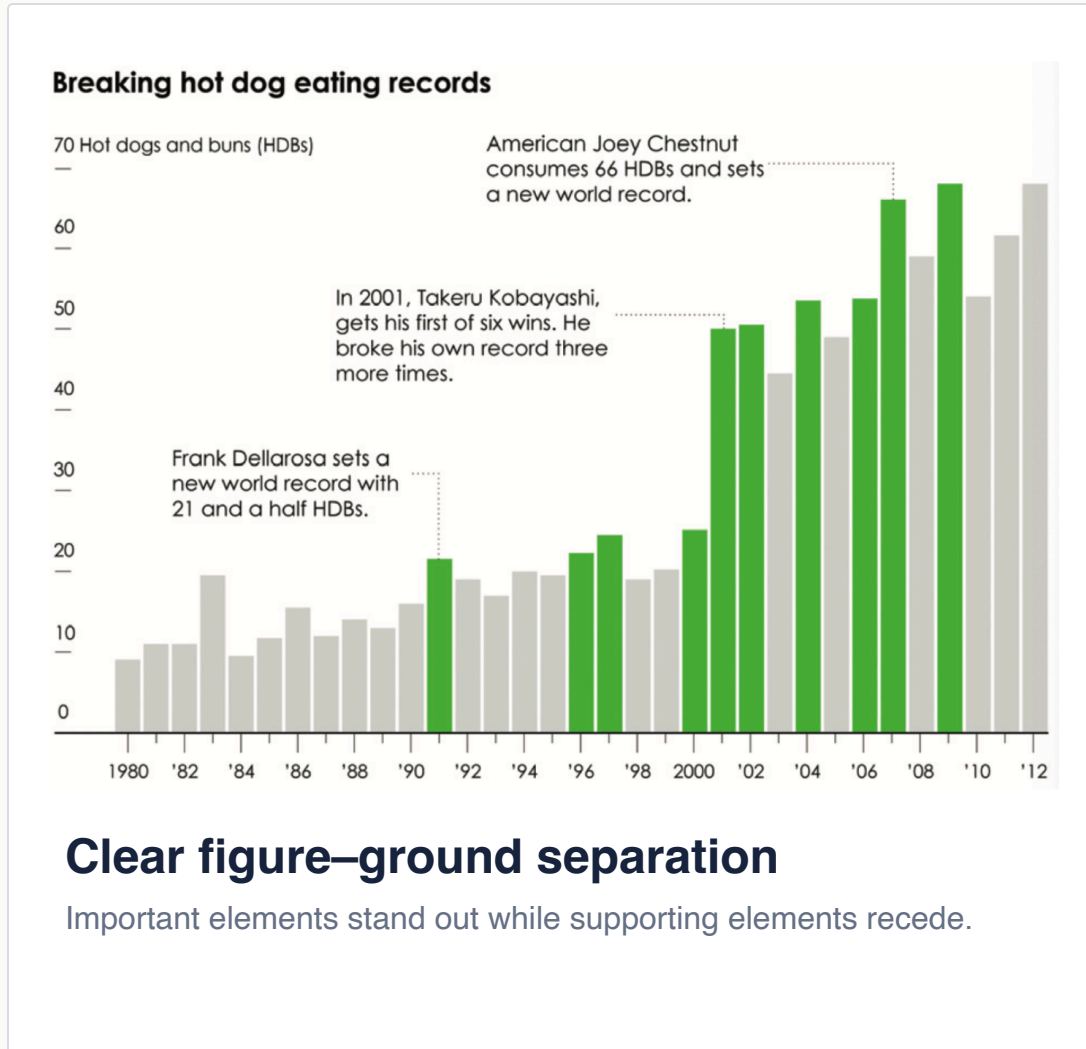
**Kanizsa triangle**



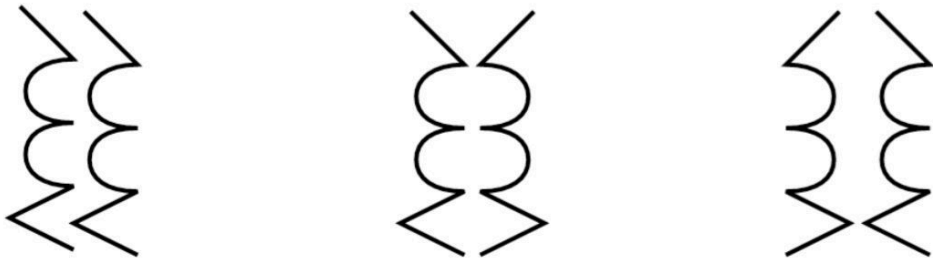
**The visual system fills in missing boundaries to perceive complete objects.**



# Figure-ground determines what stands out

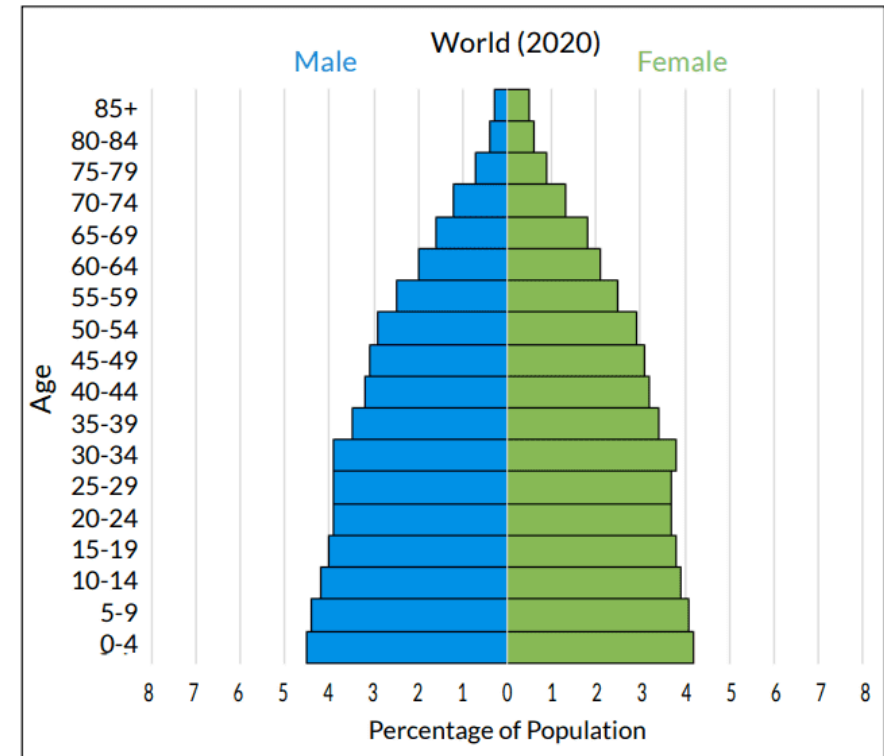


# Symmetry makes elements feel related



## Symmetry creates a visual whole.

The centre and right pairs group more strongly than the parallel lines on the left.



## Mirrored layouts support comparison.

Population pyramids use symmetry to compare two sides.

# Common fate groups elements that move together



**Elements moving in the same direction are perceived as a group.**



**Shared direction implies shared behavior.**

Motion and orientation can reveal a group even when its elements are separated.

# Gestalt laws: three organizing questions

## 01 · GROUPS

### How elements form groups

Which elements are perceived as belonging together?

Proximity

Similarity

Connectedness

Enclosure

## 02 · OBJECTS

### How the mind forms coherent objects

How does incomplete or complex visual evidence become an object?

#### Continuity

Follow smooth paths across fragments.



#### Closure

Complete objects despite missing boundaries.



#### Figure–Ground

Separate the object from its background.

## 03 · OTHER CUES

### Additional grouping cues

What else makes elements feel related?

Symmetry

Common Fate

# Which layout communicates six groups?



Vote first · Then justify with a Gestalt principle

# Which layout treats three reports equally?



Discuss in pairs · Name the cue that creates hierarchy

Part 03

# Turn principles into decisions

The goal is not to decorate—it is to control what viewers notice and infer.

# From perception to design

| What we learned           | Design implication                             |
|---------------------------|------------------------------------------------|
| Attention is selective    | Make important information stand out.          |
| Proximity / Similarity    | Make related elements appear related.          |
| Connectedness / Enclosure | Make relationships and groups explicit.        |
| Continuity                | Make paths easy to follow.                     |
| Figure–Ground             | Keep important elements in the foreground.     |
| Prior knowledge           | Design for the viewer’s task and expectations. |



# Design the viewer's perceptual path

01

## Notice

What attracts attention first?

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Selective attention  
Bottom-up and top-down processing

02

## Group

What elements appear to belong together?

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Gestalt grouping principles

03

## Follow

What relationships or paths are easy to trace?

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Continuity  
Connectedness

04

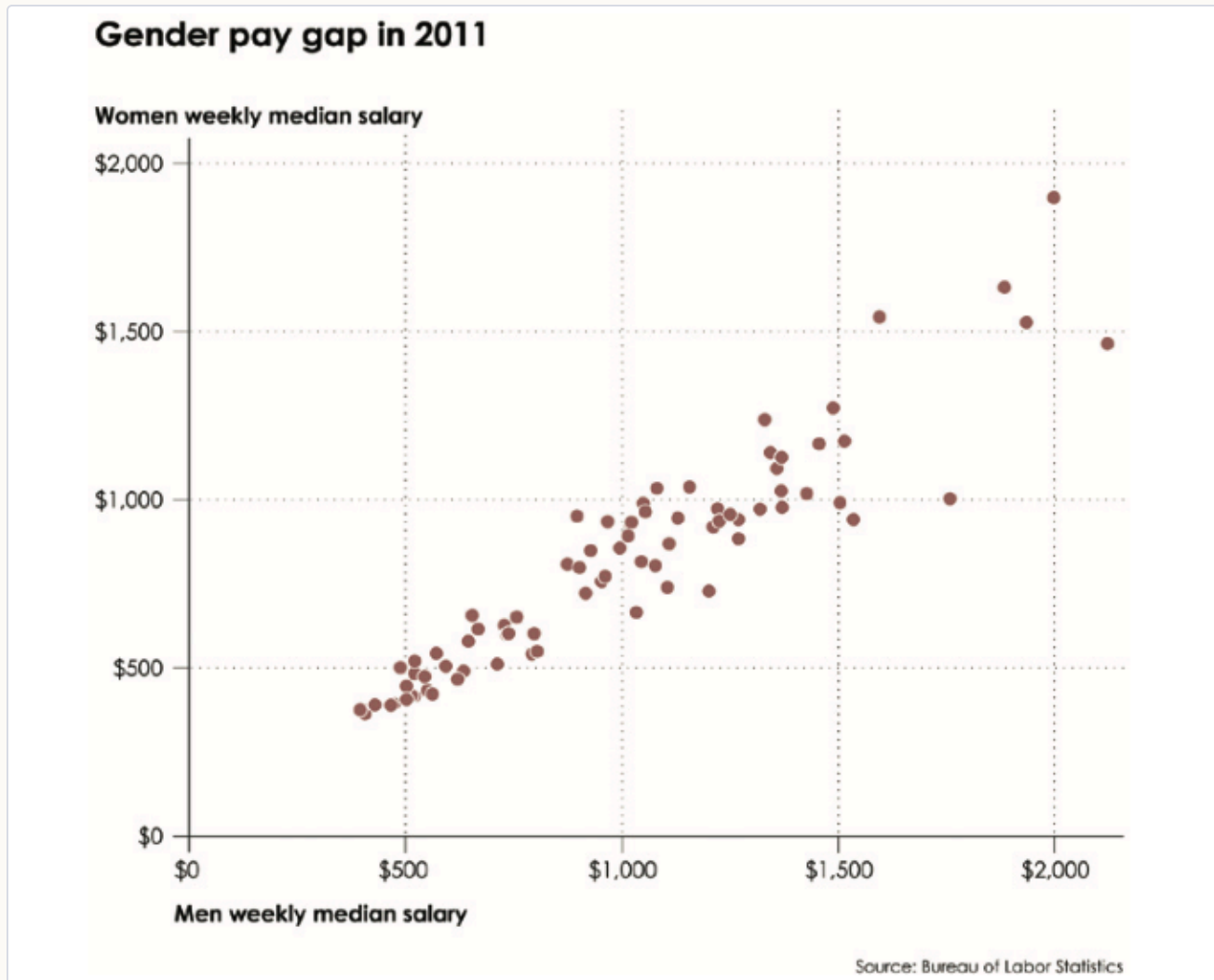
## Interpret

Does the perceived structure support the intended meaning?

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Prior knowledge  
Visual cognition

# Diagnose this visualization



## 1 · Notice

What do you notice first?

## 2 · Group

What appears grouped?

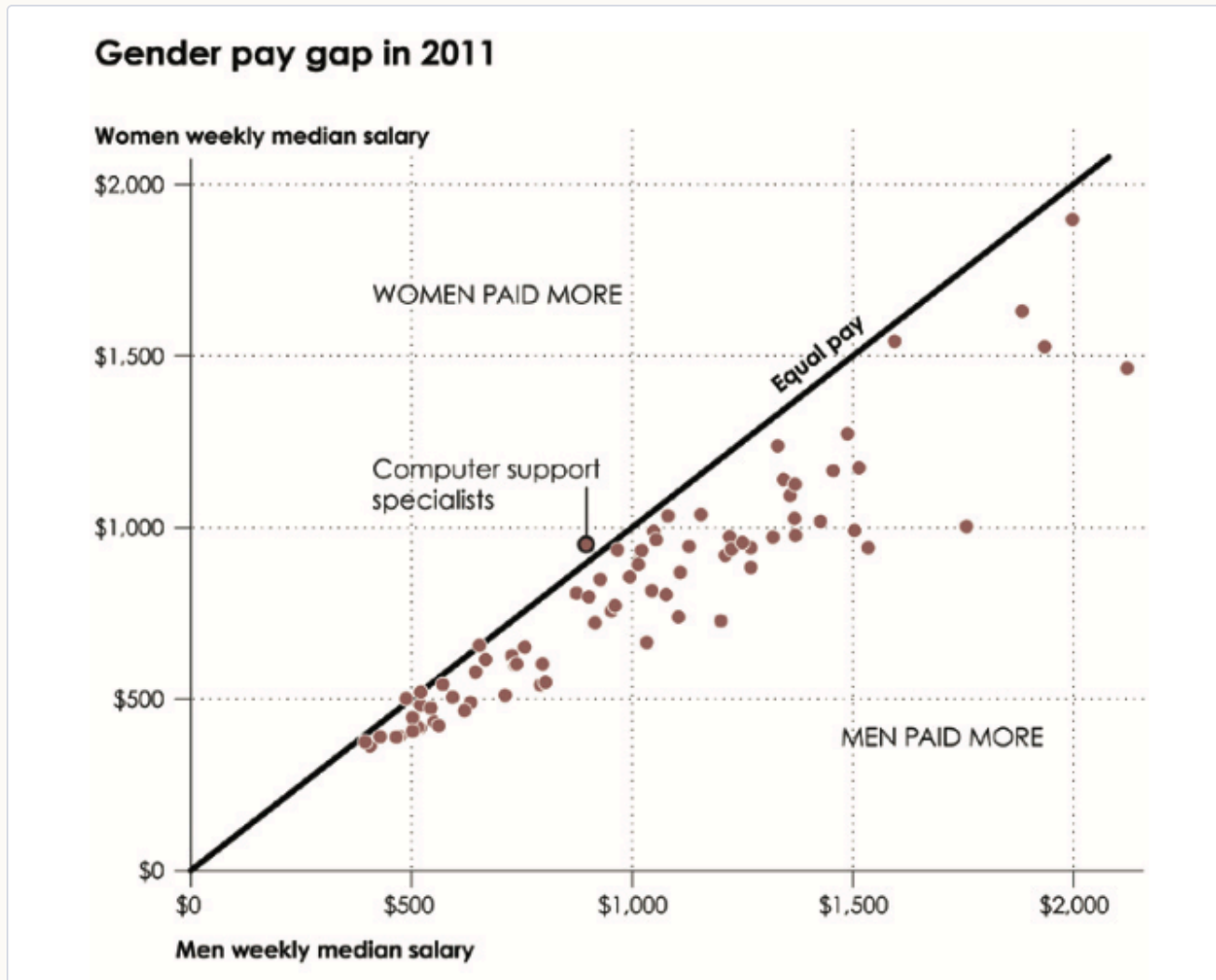
## 3 · Follow

What paths or relationships do you follow?

## 4 · Interpret

Does this support the intended interpretation?

# Diagnose this visualization



## 1 · Notice

What do you notice first?

## 2 · Group

What appears grouped?

## 3 · Follow

What paths or relationships do you follow?

## 4 · Interpret

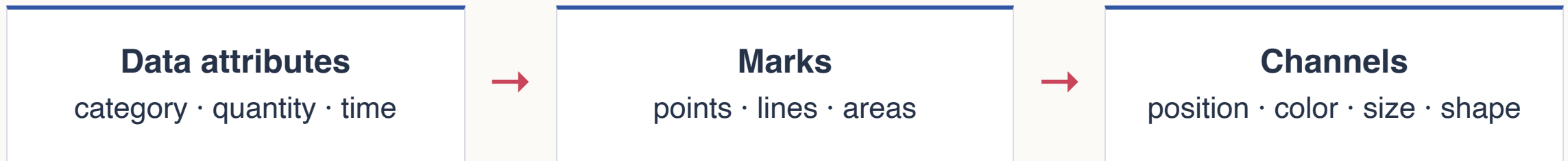
Does this support the intended interpretation?

Preview

# How does data become visible?

The next two lectures move from the human visual system to the designer's visual vocabulary.

# Visual encoding maps data to marks and channels



Today: why the viewer perceives patterns. Next: how the designer creates them.

# Three ideas to carry forward

1. Seeing is **selective**, not passive.
2. Perception is shaped by both **visual evidence** and **prior knowledge**.
3. Layout creates meaning through **grouping, separation, and hierarchy**.